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Overview

Areas of Interest. Multimodal LLMs, Post-training for Large Models, Multimodal Emotion and Social AI, Speech and Audio, Video Generation, Vision-Language Models

Summary. My research at USC aims to develop *multimodal social understanding* and *behavior generation* algorithms using *multimodal large language models* and large *diffusion models* respectively. Prior to USC, I have three years of industry experience on *speech understanding* and *multimodal content retrieval*.

Education

2024 - 2028 **PhD in Computer Science**, *University of Southern California, Los Angeles*.
GPA 4.0/4.0, Advisor - Prof. Mohammad Soleymani, Expected Graduation - 06/2028

2024 - 2026 **MS in Computer Science**, *University of Southern California, Los Angeles*.
GPA 4.0/4.0

2017 - 2021 **BS in Computer Science**, *Indian Institute of Technology, Roorkee*.
GPA 9.718/10.0 - **Third-highest GPA** amongst all the graduating students of IIT Roorkee 2021

Research Experience

May - Aug 2026 **Research Scientist Intern**, Meta Superintelligence Labs, [Full-time].
Host - Jinxi Guo
- Working on agentic systems to improve the temporal grounding and reasoning of multimodal LLMs specifically for long-context audios.

Aug 2024 - Apr 2026 **Graduate Researcher**, Institute for Creative Technologies, University of Southern California, [Part-time].
Advisor - Prof. Mohammad Soleymani
- Developed a new omni LLM preference optimization technique (**MoD-DPO**) to mitigate cross-modal hallucinations in omni LLMs – **relative improvements of 6-30%** on different cross-modal hallucination benchmarks. [CVPR 2026]
- Proposed **AVEm-DPO** for post-training multimodal LLMs (MLLMs) using preference optimization to enhance their emotion reasoning capabilities – performance **improvement of 6-19%** over different emotion benchmarks relative to the reference models. [ICLR 2026]
- Created **LibreFace-2.0**, an enhanced facial analysis toolkit with **diffusion-based synthetic data generation pipeline** to improve out-of-domain facial analysis – performance **improvement of 4-7%** over out-of-domain facial action unit benchmarks with ∼**20% reduced size**. [FG 2026]
- Worked on **Face-LLaVA**, a general vision-language model for different face analysis and face reasoning tasks, **outperforming all open source SOTA VLMs** on nine face analysis benchmarks. [WACV 2026]
- Evaluated **vision-language alignment** in MLLMs testing factual information retrieval and showed that probing internal states of the language model can reveal mis-alignment with **near 100% accuracy**. [EMNLP 2025]
- Proposed diffusion-based photorealistic listener behaviour animation using audio-visual speaker signals - **improvements of upto 73% on photorealism and 6% on motion generation**. [ICCV 2025]

Apr 2023 - Jul 2024 **Founding Research Engineer**, Anoki Inc, [Full-time].
Advisor - Dr. Susmita Ghose
- Worked on multimodal video retrieval using text, image, and audio – proposed system achieves **near 100% retrieval performance** with a much lighter framework compared to baselines [WACV 2025] [US Patent 1 2 3]

Jul 2021 - Mar 2023 **Data Scientist**, LG Ad Solutions (formerly Alphonso Inc.), [Full-time].
Advisor - Dr. Susmita Ghose
- Proposed a novel relation network-based pipeline for end-to-end speaker recognition, improving the baseline performance by **up to 12% relatively**. [Interspeech 2022] [ASRU 2023]

May - Jul 2020 **Research Intern**, Big-data Experience Lab, Adobe Research, [Full-time].
Advisor - Dr. Sumit Shekhar
- Used reinforcement learning (deep Q-learning) to learn an optimal acquisition function for active learning by modeling the active learning cycle as a Markov Decision Process. [CVPR 2022 Workshops]
- Reduced the annotation effort by using a weak learning setting where the annotator just has to verify the current model predictions on acquired samples. [US Patent App. 17/170,307]

Jan 2019 - Mar 2020 **Research Intern**, Indian Institute of Science (IISc.), Bengaluru | Indian Institute of Technology, Roorkee, [Part-time].
Advisors - Prof. R Venkatesh Babu | Prof. R. Balasubramanian
- — Worked on multi-person human pose prediction using synthetic dual person dataset to mitigate data limitations.
- — Worked on automatic evaluation of text-to-speech (TTS) systems and proposed a GAN-decoder-based scoring mechanism. [ACPR 2019]
- — Experimented with state-of-the-art universal adversarial perturbation techniques (attacks and defenses) and wrote a survey paper with over 50 citations. [Survey Paper]

Publications

** denotes equal contribution*

2026 **Do Audio LLMs Listen or Read ? Analyzing and Mitigating Paralinguistic Failures with VoxParadox**.
Jiacheng Pang*, **Ashutosh Chaubey***, Mohammad Soleymani
International Conference on Machine Learning **(ICML) 2026** [Paper] [Webpage]

2026 **MoD-DPO : Towards Mitigating Cross-modal Hallucinations in Omni LLMs using Modality Decoupled Preference Optimization**.
Ashutosh Chaubey, Jiacheng Pang, Mohammad Soleymani
Conference on Computer Vision and Pattern Recognition **(CVPR) 2026** [Paper] [Webpage]

2026 **AVERE : Improving Audiovisual Emotion Reasoning using Preference Optimization**.
Ashutosh Chaubey, Jiacheng Pang, Maksim Siniukov, Mohammad Soleymani
International Conference on Learning Representations **(ICLR) 2026** [Paper] [Webpage]

2026 **LibreFace 2.0 : Leveraging Large-Scale Synthetic Data for Fair and Generalizable Facial Analysis**.
Xulang Guan*, **Ashutosh Chaubey***, Maksim Siniukov, Belle Hsieh, Zongjian Li, Mohammad Soleymani
International Conference on Automatic Face and Gesture Recognition **(FG) 2026**

2026 **Face-LLaVA : Facial Expression and Attribute Understanding through Instruction Tuning**.
Ashutosh Chaubey, Xulang Guan, Mohammad Soleymani
Winter Conference on Applications of Computer Vision **(WACV) 2026** - R1 (6.4% accept.) [Preprint] [Webpage]

2025 **Can VLMs Recall Factual Associations From Visual References ?**.
Dhananjay Ashok, **Ashutosh Chaubey**, Hirona Arai, Jonathan May, Jesse Thomason
Conference on Empirical Methods in Natural Language Processing **(EMNLP) 2025** [Paper]

2025 **DiTaiListener : Controllable High Fidelity Listener Video Generation with Diffusion**.
Maksim Siniukov, Di Chang, Minh Tran, Hongkun Gong, **Ashutosh Chaubey**, Mohammad Soleymani
International Conference on Computer Vision **(ICCV) 2025** [Paper] [Webpage]

2024 **ContextIQ : A Multimodal Expert-Based Video Retrieval System for Contextual Advertising**.
Ashutosh Chaubey, Anoubhav Agarwaal, Sartaki Roy, Aayush Agrawal, Susmita Ghose
Winter Conference on Applications of Computer Vision **(WACV) 2025** [Paper] [Poster]

2023 **Meta-Learning Framework for End-to-End Imposter Identification in Unseen Speaker Recognition**.
Ashutosh Chaubey, Sparsh Sinha, Susmita Ghose
IEEE Workshop on Automatic Speech and Understanding **(ASRU) 2023** [Paper] [Poster]

2022 **Improved Relation Networks for End-to-End Speaker Verification and Identification**.
Ashutosh Chaubey, Sparsh Sinha, Susmita Ghose
Interspeech 2022 [Paper] [Poster]

2020 **Universal Adversarial Perturbations : A Survey**.
Ashutosh Chaubey*, Nikhil Agrawal*, Kavya Barnwal, Keerat K. Guliani, Pramod Mehta
arXiv Preprint (**50+ citations**) [Paper]

Awards and Recognition

2026 Recognized as a **Gold Reviewer** at ICML 2026 for outstanding contribution to community.

2026 Received the CVPR 2026 Travel Scholarship to present my paper – MoD-DPO.

2024 Received the USC Viterbi Fellowship to support my PhD at USC.

2021 Received Institute Bronze Medal at IIT Roorkee for having third highest GPA among all graduating students.

2017 All India Rank 402 among 200k students in JEE-Advanced for getting admission into IITs.

Academic Services

Conference Reviewer: NeurIPS 2026, BMVC 2026, ECCV 2026, ICML 2026, CVPR 2026, FG 2026, ICCV 2025

Teaching Assistant: CS561 (Foundations of AI) – Fall 2025, CSCI 535 (Multimodal Probabilistic Learning of Human Communication) – Spring 2025

Supervised Students

Jiacheng Pang – Research Intern, MS Student at USC (Now, PhD Student at USC), Summer - Fall 2025
Xulang Guan – CURVE Fellow, Undergraduate at USC, Fall 2024 - Fall 2025
Belle Hsieh – Research Intern, Undergraduate at UPenn, Summer 2025
Hongkun Gong – CURVE Fellow, Undergraduate at USC (Now, MS Student at Columbia), Fall 2024

Skills

Coding Languages - Python [Advanced], C++ [Intermediate]
Frameworks/Libraries - PyTorch, NumPy, Pandas, Transformers
Tools - VSCode, Git, Anaconda, Docker